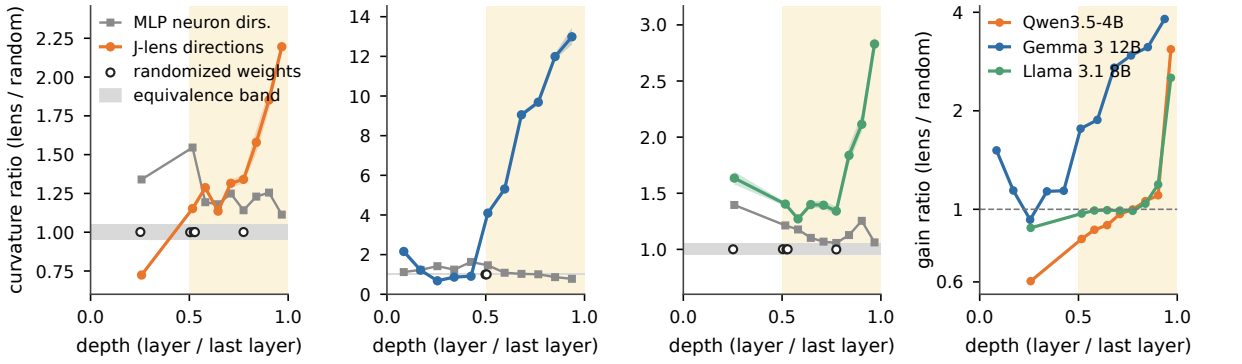
(a) Qwen3.5-4B
(b) Gemma 3 12B
(c) Llama 3.1 8B
(d) First-order gain |J v|
MLP neuron dirs.
J-lens directions
randomized weights
equivalence band
Qwen3.5-4B
Gemma 3 12B
Llama 3.1 8B
curvature ratio (lens / random)
gain ratio (lens / random)
depth (layer / last layer)
0.0
0.5
1.0
0.75
1.00
1.25
1.50
1.75
2.00
2.25
0
2
4
6
8
10
12
14
1.0
1.5
2.0
2.5
3.0
0.6
1
2
4